

EM Algorithm

1.1 Overview

Suppose that the likelihood of the observed data Y_{obs} is difficult to maximize. However, given additional unobserved data Y_{miss} , the likelihood maximization becomes tractable. Let $\ln f(Y_{\text{obs}}, Y_{\text{miss}}|\boldsymbol{\theta})$ denote the complete data log likelihood, and $\ln f(Y_{\text{obs}}|\boldsymbol{\theta})$ the observed data log likelihood. The Expectation-Maximization algorithm proceeds as follows:

1. **E-step:** Given a current estimate of the parameter $\boldsymbol{\theta}^{(r)}$, calculate the expectation of the complete data log likelihood given the observed data:

$$Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}) = \int \ln f(Y_{\text{obs}}, Y_{\text{miss}}|\boldsymbol{\theta}) f(Y_{\text{miss}}|Y_{\text{obs}}, \boldsymbol{\theta}^{(r)}) dY_{\text{miss}}.$$

2. **M-step:** Maximize the current objective function to update the estimate of $\boldsymbol{\theta}$:

$$\boldsymbol{\theta}^{(r+1)} \leftarrow \arg \max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)})$$

Remark 1.1. Provided differentiation in $\boldsymbol{\theta}$ commutes with integration over $f(Y_{\text{miss}}|Y_{\text{obs}}, \boldsymbol{\theta})$, the EM-algorithm may be implemented as follows:

- Derive the complete data score equations $\ell(\boldsymbol{\theta})$:

$$\mathcal{U}(\boldsymbol{\theta}) = \frac{\partial}{\partial \boldsymbol{\theta}} \ln f(Y_{\text{obs}}, Y_{\text{miss}}|\boldsymbol{\theta}).$$

- Take the expectation of the complete data score equations given Y_{obs} and $\boldsymbol{\theta}^{(r)}$:

$$\mathcal{U}(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}) = \mathbb{E} \{ \mathcal{U}(\boldsymbol{\theta}) | Y_{\text{obs}}, \boldsymbol{\theta}^{(r)} \}.$$

- Obtain $\boldsymbol{\theta}^{(r+1)}$ by solving:

$$\mathcal{U}(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}) \stackrel{\text{Set}}{=} \mathbf{0}.$$

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Remark 1.2. The overall EM algorithm converges linearly, regardless of the maximization procedure employed in the M-step. Thus, stable linear optimization methods, such as coordinate ascent, are perhaps preferable to faster yet less stable optimization methods, such as Newton-Raphson.

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1.2 Information Inequality

Definition 1.1. The **cross entropy** between densities f and g is:

$$\mathcal{H}(f||g) = - \int \ln \{g(x)\} f(x) dx = -\mathbb{E}_f \ln \{g(x)\}.$$

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Proposition 1.1 (Gibbs' Inequality). For densities f and g :

$$\mathcal{H}(f||g) - \mathcal{H}(f||f) \geq 0. \quad (1.1)$$

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Proof.

$$\begin{aligned} \mathcal{H}(f||g) - \mathcal{H}(f||f) &= - \int \ln \{g(x)\} f(x) dx + \int \ln \{f(x)\} f(x) dx \\ &= - \int \ln \left\{ \frac{g(x)}{f(x)} \right\} f(x) dx. \end{aligned}$$

For $x \in (0, \infty)$, $\ln(x) \leq x - 1$, or $-\ln(x) \geq -(x - 1)$, therefore:

$$\begin{aligned} - \int \ln \left\{ \frac{g(x)}{f(x)} \right\} f(x) dx &\geq - \int \left\{ \frac{g(x)}{f(x)} - 1 \right\} f(x) dx \\ &\geq - \int g(x) dx + \int f(x) dx = -1 + 1 = 0. \end{aligned}$$

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Definition 1.2. The **Kullback-Leibler divergence** between f and g is:

$$\mathbb{KL}(f||g) \equiv \mathcal{H}(f||g) - \mathcal{H}(f||f) = - \int \ln \left\{ \frac{g(x)}{f(x)} \right\} f(x) dx. \quad (1.2)$$

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1.3 Verification of the EM Algorithm

Definition 1.3. The **EM objective** is defined as:

$$Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}) = \int \ln f(Y_{\text{obs}}, Y_{\text{miss}}|\boldsymbol{\theta}) f(Y_{\text{miss}}|Y_{\text{obs}}, \boldsymbol{\theta}^{(r)}) dY_{\text{miss}}. \quad (1.3)$$

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Proposition 1.2. The observed data log likelihood decomposes as the EM objective plus the cross entropy $\mathcal{H}(\boldsymbol{\theta}||\boldsymbol{\theta}^{(r)})$ between $f(Y_{\text{miss}}|Y_{\text{obs}}; \boldsymbol{\theta}^{(r)})$ and $f(Y_{\text{miss}}|Y_{\text{obs}}; \boldsymbol{\theta})$. That is:

$$\ln f(Y_{\text{obs}}|\boldsymbol{\theta}) = Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}) + \mathcal{H}(\boldsymbol{\theta}||\boldsymbol{\theta}^{(r)}), \quad (1.4)$$

where:

$$\mathcal{H}(\boldsymbol{\theta}||\boldsymbol{\theta}^{(r)}) = - \int \ln \{f(Y_{\text{miss}}|Y_{\text{obs}}; \boldsymbol{\theta})\} f(Y_{\text{miss}}|Y_{\text{obs}}; \boldsymbol{\theta}^{(r)}) dY_{\text{miss}}.$$

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Proof. The conditional density of the missing data given the observed is:

$$f(Y_{\text{miss}}|Y_{\text{obs}}; \boldsymbol{\theta}) = \frac{f(Y_{\text{obs}}, Y_{\text{miss}}|\boldsymbol{\theta})}{f(Y_{\text{obs}}|\boldsymbol{\theta})}$$

Upon taking the logarithm:

$$\ln f(Y_{\text{obs}}|\boldsymbol{\theta}) = \ln f(Y_{\text{obs}}, Y_{\text{miss}}|\boldsymbol{\theta}) - \ln f(Y_{\text{miss}}|Y_{\text{obs}}; \boldsymbol{\theta}).$$

Taking the expectation with respect to the density $f(Y_{\text{miss}}|Y_{\text{obs}}; \boldsymbol{\theta}^{(r)})$:

$$\begin{aligned} \ln f(Y_{\text{obs}}|\boldsymbol{\theta}) &= \int \ln \{f(Y_{\text{obs}}, Y_{\text{miss}}|\boldsymbol{\theta})\} f(Y_{\text{miss}}|Y_{\text{obs}}; \boldsymbol{\theta}^{(r)}) dY_{\text{miss}} \\ &\quad - \int \ln \{f(Y_{\text{miss}}|Y_{\text{obs}}; \boldsymbol{\theta})\} f(Y_{\text{miss}}|Y_{\text{obs}}; \boldsymbol{\theta}^{(r)}) dY_{\text{miss}} \\ &= Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}) + \mathcal{H}(\boldsymbol{\theta}||\boldsymbol{\theta}^{(r)}). \end{aligned}$$

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Proposition 1.3 (Increment Property). Increasing the EM objective causes an increase at least as great in the observed data log likelihood:

$$\ln f(Y_{\text{obs}}|\boldsymbol{\theta}) - \ln f(Y_{\text{obs}}|\boldsymbol{\theta}^{(r)}) \geq Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}) - Q(\boldsymbol{\theta}^{(r)}|\boldsymbol{\theta}^{(r)}).$$

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Proof. Substituting $\boldsymbol{\theta}^{(r)}$ for $\boldsymbol{\theta}$ in (1.4):

$$\ln f(Y_{\text{obs}}|\boldsymbol{\theta}^{(r)}) = Q(\boldsymbol{\theta}^{(r)}|\boldsymbol{\theta}^{(r)}) + \mathcal{H}(\boldsymbol{\theta}^{(r)}||\boldsymbol{\theta}^{(r)}). \quad (1.5)$$

Subtracting (1.5) from (1.4):

$$\ln f(Y_{\text{obs}}|\boldsymbol{\theta}) - \ln f(Y_{\text{obs}}|\boldsymbol{\theta}^{(r)}) = Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}) - Q(\boldsymbol{\theta}^{(r)}|\boldsymbol{\theta}^{(r)}) + \mathcal{H}(\boldsymbol{\theta}||\boldsymbol{\theta}^{(r)}) - \mathcal{H}(\boldsymbol{\theta}^{(r)}||\boldsymbol{\theta}^{(r)}).$$

From Gibbs' inequality (1.1), $\mathcal{H}(\boldsymbol{\theta}||\boldsymbol{\theta}^{(r)}) - \mathcal{H}(\boldsymbol{\theta}^{(r)}||\boldsymbol{\theta}^{(r)}) \geq 0$, therefore:

$$\ln f(Y_{\text{obs}}|\boldsymbol{\theta}) - \ln f(Y_{\text{obs}}|\boldsymbol{\theta}^{(r)}) \geq Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}) - Q(\boldsymbol{\theta}^{(r)}|\boldsymbol{\theta}^{(r)}).$$

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1.4 Connection to Variational Inference

Proposition 1.4. The observed data log likelihood is expressible as:

$$\ln f(Y_{\text{obs}}|\boldsymbol{\theta}) = \mathcal{L}\{\boldsymbol{\theta}, g(Y_{\text{miss}})\} + \mathbb{KL}\{g(Y_{\text{miss}})||f(Y_{\text{miss}}|Y_{\text{obs}};\boldsymbol{\theta})\},$$

where $g(Y_{\text{miss}})$ is any density over Y_{miss} ,

$$\mathcal{L}\{\boldsymbol{\theta}, g(Y_{\text{miss}})\} = \int \ln \left\{ \frac{f(Y_{\text{obs}}, Y_{\text{miss}}|\boldsymbol{\theta})}{g(Y_{\text{miss}})} \right\} g(Y_{\text{miss}}) dY_{\text{miss}}.$$

is the **evidence lower bound** (ELBO), and:

$$\mathbb{KL}\{g(Y_{\text{miss}})||f(Y_{\text{miss}}|Y_{\text{obs}};\boldsymbol{\theta})\} = - \int \ln \left\{ \frac{f(Y_{\text{miss}}|Y_{\text{obs}};\boldsymbol{\theta})}{g(Y_{\text{miss}})} \right\} g(Y_{\text{miss}}) dY_{\text{miss}}$$

is the Kullback-Leibler divergence (1.2) between $g(Y_{\text{miss}})$ and $f(Y_{\text{miss}}|Y_{\text{obs}};\boldsymbol{\theta})$. ◆

Proof. Recall from before:

$$\ln f(Y_{\text{obs}}|\boldsymbol{\theta}) = \ln f(Y_{\text{obs}}, Y_{\text{miss}}|\boldsymbol{\theta}) - \ln f(Y_{\text{miss}}|Y_{\text{obs}};\boldsymbol{\theta}).$$

Let $g(Y_{\text{miss}})$ denote some density on Y_{miss} . Adding and subtracting $\ln\{g(Y_{\text{miss}})\}$:

$$\begin{aligned} \ln f(Y_{\text{obs}}|\boldsymbol{\theta}) &= \{\ln f(Y_{\text{obs}}, Y_{\text{miss}}|\boldsymbol{\theta}) - \ln g(Y_{\text{miss}})\} - \{\ln f(Y_{\text{miss}}|Y_{\text{obs}};\boldsymbol{\theta}) - \ln g(Y_{\text{miss}})\} \\ &= \ln \left\{ \frac{f(Y_{\text{obs}}, Y_{\text{miss}}|\boldsymbol{\theta})}{g(Y_{\text{miss}})} \right\} - \ln \left\{ \frac{f(Y_{\text{miss}}|Y_{\text{obs}};\boldsymbol{\theta})}{g(Y_{\text{miss}})} \right\}. \end{aligned}$$

Taking the expectation with respect to $g(Y_{\text{miss}})$:

$$\begin{aligned} \ln f(Y_{\text{obs}}|\boldsymbol{\theta}) &= \int \ln \left\{ \frac{f(Y_{\text{obs}}, Y_{\text{miss}}|\boldsymbol{\theta})}{g(Y_{\text{miss}})} \right\} g(Y_{\text{miss}}) dY_{\text{miss}} \\ &\quad - \int \ln \left\{ \frac{f(Y_{\text{miss}}|Y_{\text{obs}};\boldsymbol{\theta})}{g(Y_{\text{miss}})} \right\} g(Y_{\text{miss}}) dY_{\text{miss}}. \end{aligned}$$

The first term is the evidence lower bound:

$$\mathcal{L}\{\boldsymbol{\theta}, g(Y_{\text{miss}})\} = \int \ln \left\{ \frac{f(Y_{\text{obs}}, Y_{\text{miss}}|\boldsymbol{\theta})}{g(Y_{\text{miss}})} \right\} g(Y_{\text{miss}}) dY_{\text{miss}}.$$

The second term is the Kullback-Leibler divergence between $g(Y_{\text{miss}})$ and $f(Y_{\text{miss}}|Y_{\text{obs}};\boldsymbol{\theta})$:

$$\mathbb{KL}\{g(Y_{\text{miss}})||f(Y_{\text{miss}}|Y_{\text{obs}};\boldsymbol{\theta})\} = - \int \ln \left\{ \frac{f(Y_{\text{miss}}|Y_{\text{obs}};\boldsymbol{\theta})}{g(Y_{\text{miss}})} \right\} g(Y_{\text{miss}}) dY_{\text{miss}}.$$

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Corollary 1.1. Since, from Gibbs' inequality (1.1), $\mathbb{KL}(\cdot||\cdot) \geq 0$, the observed data log likelihood is not less than the evidence lower bound:

$$\ln f(Y_{\text{obs}}|\boldsymbol{\theta}) \geq \mathcal{L}\{\boldsymbol{\theta}, g(Y_{\text{miss}})\}.$$



Corollary 1.2. Substituting $f(Y_{\text{miss}}|Y_{\text{obs}}, \boldsymbol{\theta}^{(r)})$ for $g(Y_{\text{miss}})$ in the expression for the evidence lower bound gives:

$$\mathcal{L}\{\boldsymbol{\theta}, f(Y_{\text{miss}}|Y_{\text{obs}}, \boldsymbol{\theta}^{(r)})\} = Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}) + \mathcal{H}\{\boldsymbol{\theta}^{(r)}||\boldsymbol{\theta}^{(r)}\}.$$



Proof.

$$\begin{aligned} \mathcal{L}\{\boldsymbol{\theta}, f(Y_{\text{miss}}|Y_{\text{obs}}, \boldsymbol{\theta}^{(r)})\} &= \int \ln \left\{ \frac{f(Y_{\text{obs}}, Y_{\text{miss}}|\boldsymbol{\theta})}{f(Y_{\text{miss}}|Y_{\text{obs}}, \boldsymbol{\theta}^{(r)})} \right\} f(Y_{\text{miss}}|Y_{\text{obs}}, \boldsymbol{\theta}^{(r)}) dY_{\text{miss}} \\ &= \int \ln \{f(Y_{\text{obs}}, Y_{\text{miss}}|\boldsymbol{\theta})\} f(Y_{\text{miss}}|Y_{\text{obs}}, \boldsymbol{\theta}^{(r)}) dY_{\text{miss}} \\ &\quad - \int \ln \{f(Y_{\text{miss}}|Y_{\text{obs}}, \boldsymbol{\theta}^{(r)})\} f(Y_{\text{miss}}|Y_{\text{obs}}, \boldsymbol{\theta}^{(r)}) dY_{\text{miss}} \\ &= Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}) + \mathcal{H}\{\boldsymbol{\theta}^{(r)}||\boldsymbol{\theta}^{(r)}\}. \end{aligned}$$



1.5 Information Matrix

Definition 1.4. The **complete data score** is:

$$\mathcal{U}(\boldsymbol{\theta}) = \frac{\partial}{\partial \boldsymbol{\theta}} \ln f(Y_{\text{obs}}, Y_{\text{miss}}|\boldsymbol{\theta}).$$

The **EM score** is:

$$\mathcal{U}(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}) = \frac{\partial}{\partial \boldsymbol{\theta}} Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}). \quad (1.6)$$



Proposition 1.5. If differentiation in $\boldsymbol{\theta}$ commutes with integration over $f(Y_{\text{miss}}|Y_{\text{obs}}, \boldsymbol{\theta})$, then:

$$\mathbb{E}\{\mathcal{U}(\boldsymbol{\theta})|Y_{\text{obs}}, \boldsymbol{\theta}^{(r)}\} = \frac{\partial}{\partial \boldsymbol{\theta}} Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}) = \mathcal{U}(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}).$$

That is, the EM score is the expectation of the complete data score given $(Y_{\text{obs}}, \boldsymbol{\theta}^{(r)})$. 

Proof.

$$\begin{aligned}\mathbb{E}\{\mathcal{U}(\boldsymbol{\theta})|Y_{\text{obs}}, \boldsymbol{\theta}^{(r)}\} &= \mathbb{E}\left\{\frac{\partial}{\partial \boldsymbol{\theta}} \ln f(Y_{\text{obs}}, Y_{\text{miss}}|\boldsymbol{\theta})|Y_{\text{obs}}, \boldsymbol{\theta}^{(r)}\right\} \\ &= \frac{\partial}{\partial \boldsymbol{\theta}} \mathbb{E}\{\ln f(Y_{\text{obs}}, Y_{\text{miss}}|\boldsymbol{\theta})|Y_{\text{obs}}, \boldsymbol{\theta}^{(r)}\} = \frac{\partial}{\partial \boldsymbol{\theta}} Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}).\end{aligned}$$

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Definition 1.5. The **complete data expected information** is:

$$\mathcal{I}(\boldsymbol{\theta}) = \mathbb{E}\{\mathcal{U}(\boldsymbol{\theta}) \otimes \mathcal{U}(\boldsymbol{\theta})\}.$$

The **EM expected information** is the variance of the EM score:

$$\mathcal{I}(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}) = \mathbb{V}\{\mathcal{U}(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)})\}.$$

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Proposition 1.6 (Total Variance Decomposition).

$$\mathcal{I}(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}) = \mathcal{I}(\boldsymbol{\theta}) - \mathbb{E}\left[\mathbb{V}\{\mathcal{U}(\boldsymbol{\theta})|Y_{\text{obs}}, \boldsymbol{\theta}^{(r)}\}\right]. \quad (1.7)$$

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Proof. Observe that:

$$\mathcal{I}(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}) = \mathbb{V}\{\mathcal{U}(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)})\} = \mathbb{V}\left[\mathbb{E}\{\mathcal{U}(\boldsymbol{\theta})|Y_{\text{obs}}, \boldsymbol{\theta}^{(r)}\}\right].$$

By law of total variance:

$$\begin{aligned}\mathcal{I}(\boldsymbol{\theta}) &= \mathbb{V}\{\mathcal{U}(\boldsymbol{\theta})\} \\ &= \mathbb{E}\left[\mathbb{V}\{\mathcal{U}(\boldsymbol{\theta})|Y_{\text{obs}}, \boldsymbol{\theta}^{(r)}\}\right] + \mathbb{V}\left[\mathbb{E}\{\mathcal{U}(\boldsymbol{\theta})|Y_{\text{obs}}, \boldsymbol{\theta}^{(r)}\}\right] \\ &= \mathbb{E}\left[\mathbb{V}\{\mathcal{U}(\boldsymbol{\theta})|Y_{\text{obs}}, \boldsymbol{\theta}^{(r)}\}\right] + \mathcal{I}(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}).\end{aligned}$$

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Remark 1.3. The EM information may be found either by:

1. Directly finding the variance of the EM score in (1.6).
2. Applying the total variance decomposition in (1.7).

To obtain a final estimate of the information for use in inference, $\mathcal{I}(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)})$ will be evaluated at $\boldsymbol{\theta} = \boldsymbol{\theta}^{(r)}$, so the distinction between $\boldsymbol{\theta}$ and $\boldsymbol{\theta}^{(r)}$ may be dropped. ◆

Extensions

2.1 Monte Carlo EM

Monte Carlo (MC) EM is useful when the expectation in the E-step is intractable, but it is possible to simulate from $f(Y_{\text{miss}}|Y_{\text{obs}}; \boldsymbol{\theta})$. In MC-EM, on the E-step, a sample $(Y_{\text{miss}}^{(1)}, \dots, Y_{\text{miss}}^{(M)})$ of size M is drawn from $f(Y_{\text{miss}}|Y_{\text{obs}}; \boldsymbol{\theta}^{(r)})$, and the EM objective (1.3) is approximated by:

$$\hat{Q}(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}) = \frac{1}{M} \sum_{m=1}^M \ln f(Y_{\text{obs}}, Y_{\text{miss}}^{(m)}|\boldsymbol{\theta}).$$

Similarly, the score may be approximated by:

$$\hat{\mathcal{U}}(\boldsymbol{\theta}) = \frac{1}{M} \sum_{m=1}^M \dot{\ell}(Y_{\text{obs}}, Y_{\text{miss}}^{(m)}|\boldsymbol{\theta}),$$

where:

$$\dot{\ell}(Y_{\text{obs}}, Y_{\text{miss}}^{(m)}|\boldsymbol{\theta}) = \frac{\partial}{\partial \boldsymbol{\theta}} \ln f(Y_{\text{obs}}, Y_{\text{miss}}^{(m)}|\boldsymbol{\theta}).$$

Finally, the observed information is approximated as:

$$\hat{\mathcal{J}}(\boldsymbol{\theta}) = \frac{1}{M} \sum_{m=1}^M \dot{\ell}(Y_{\text{obs}}, Y_{\text{miss}}^{(m)}|\boldsymbol{\theta}) \otimes \dot{\ell}(Y_{\text{obs}}, Y_{\text{miss}}^{(m)}|\boldsymbol{\theta}).$$

2.2 Expectation Conditional Maximization

The expectation conditional maximization (ECM) algorithm is useful when jointly maximizing $Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)})$ with respect to all of $\boldsymbol{\theta}$ is challenging; however, the maximization may be split into a sequence of simpler, conditional maximizations. For example, suppose there is a natural partition of the parameter as $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \dots, \boldsymbol{\theta}_K)$. The M-step of ECM proceeds as follows:

$$\begin{aligned} \boldsymbol{\theta}_1^{(r+1)} &\leftarrow \arg \max_{\boldsymbol{\theta}_1 \in \Theta_1} Q(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2^{(r)}, \dots, \boldsymbol{\theta}_K^{(r)}|\boldsymbol{\theta}^{(r)}) \\ \boldsymbol{\theta}_2^{(r+1)} &\leftarrow \arg \max_{\boldsymbol{\theta}_2 \in \Theta_2} Q(\boldsymbol{\theta}_1^{(r+1)}, \boldsymbol{\theta}_2, \dots, \boldsymbol{\theta}_K^{(r)}|\boldsymbol{\theta}^{(r)}) \\ &\vdots \\ \boldsymbol{\theta}_K^{(r+1)} &\leftarrow \arg \max_{\boldsymbol{\theta}_K \in \Theta_K} Q(\boldsymbol{\theta}_1^{(r+1)}, \boldsymbol{\theta}_2^{(r+1)}, \dots, \boldsymbol{\theta}_K|\boldsymbol{\theta}^{(r)}) \end{aligned}$$

Moreover, ECM maximizes only the EM objective $Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)})$ at each conditional maximization. The ECME algorithm allows some of those steps to maximize the observed data log likelihood $\ln f(Y_{\text{obs}}|\boldsymbol{\theta})$ instead. This is useful when the observed data log likelihood admits closed form solutions for some components of $\boldsymbol{\theta}$.